Vocabulary Types

The small standard types that make a function's contract visible in its signature: `pair` and `tuple`, `optional`, `variant`, `expected` and `span`. Builds on Types, Initialization & Casts.

Why vocabulary types

A signature like `int find(const char* s)` hides its contract: does `-1` mean "not found"? Is `s` allowed to be null? Who owns the result? A **vocabulary type** puts the answer in the type, so every reader and the compiler see it.

THE QUESTION	THE OLD ANSWER	THE TYPE
Two or three values back from one function?	out-parameters, a throwaway struct	`std::pair`, `std::tuple`, or a named struct
A value that may be absent?	a magic value (`-1`), a null pointer	`std::optional<T>`
One of several different types?	a `union` plus a tag, a base class	`std::variant<A, B>`
A value, or the reason there is none?	an error code plus an out-parameter, an exception	`std::expected<T, E>` (C++23)
A run of elements?	a pointer plus a count	`std::span<T>`, `std::string_view`

All of them are ordinary value types: no heap allocation, no virtual functions, nothing to free.

`pair`, `tuple` and structured bindings

`std::pair<A, B>` holds two values, `std::tuple<A, B, C ...>` any number. Structured bindings (C++17) unpack them into named variables.

```
std::pair<int, int> minmax(const std::vector<int>& v);      // returns {smallest, largest}
auto [lo, hi] = minmax({3, 9, 1, 7});                       // lo = 1, hi = 9: two names instead of .first / .second

auto t = std::make_tuple(1, 2.5, std::string("x"));
std::get<1>(t);                    // 2.5   (index must be a compile-time constant)
std::get<std::string>(t);          // "x"   (or by type, when the type appears once)

std::map<std::string, int> m{{"a", 1}};
auto [it, inserted] = m.insert({"a", 2});   // insert returns pair<iterator, bool>
// inserted == false, it->second == 1: the key existed, nothing changed
```

- A pair or tuple has **no names**: `.first` and `std::get<2>` say nothing about meaning. For a result that leaves a function and is used in more than one place, a small named `struct` reads better.
- Comparison is built in and lexicographic: pairs and tuples compare element by element, so they work as `map` keys and in `std::sort`.
- `std::tie(a, b) = f();` assigns into existing variables; `std::ignore` skips one.
- Structured bindings also unpack plain structs and arrays: `auto [x, y] = point;`. They are copies unless you write `auto&` or `const auto&`.

`optional`: a value or nothing

`std::optional<T>` holds either a `T` or nothing. The "nothing" is part of the type, so a caller cannot forget it exists.

```
std::optional<int> parseInt(std::string_view s) {
    int v = 0;
    auto r = std::from_chars(s.data(), s.data() + s.size(), v);
    if (r.ec ≠ std::errc{} || r.ptr ≠ s.data() + s.size()) return
std::nullopt;   // "no value"
    return v;
// converts to optional<int>
}

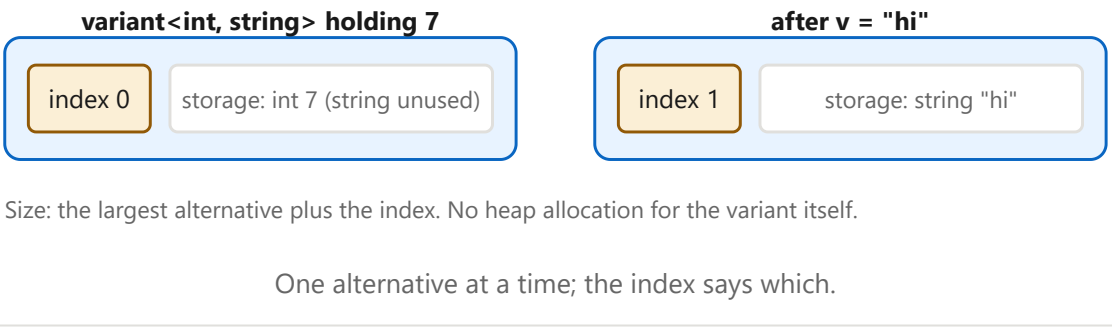
if (auto n = parseInt("42")) use(*n);        // test, then dereference: n is
true when a value is present
parseInt("4x").value_or(-1);                 // -1: a default for the empty
case
parseInt("").has_value();                    // false
std::optional<int> e;  e.value();            // throws
std::bad_optional_access
*e;                                          // undefined behaviour: no
check, like dereferencing a null pointer
```

OPERATION	ON AN EMPTY OPTIONAL
`if (o)` / `o.has_value()`	false
`*o`, `o→member`	undefined behaviour: check first
`o.value()`	throws `bad_optional_access`
`o.value_or(x)`	returns `x`
`o.reset()`, `o = std::nullopt`	makes it empty

- The value lives **inside** the optional (compiled: `sizeof(int)` is 4, `sizeof(optional<int>)` is 8: the value plus a flag, padded). No heap allocation, unlike a pointer or `unique_ptr`.
- Use it for a result that is *legitimately* absent: a lookup that may miss, a parse of untrusted input, an optional configuration value. It does not say *why* the value is missing; for that, use `expected`.
- Not a replacement for a reference or pointer that means "refers to an object owned elsewhere": `optional<T&>` does not exist. Use a pointer there.
- Marking a function `[[nodiscard]]` keeps callers from dropping the result.

`variant` and `std::visit`

`std::variant<A, B, C>` is a type-safe union: it holds exactly one of its alternatives and knows which.



Size: the largest alternative plus the index. No heap allocation for the variant itself.

One alternative at a time; the index says which.

```
std::variant<int, std::string> v = 7;
v.index();                                  // 0: the int is active
std::holds_alternative<int>(v);             // true
std::get<int>(v);                           // 7
std::get<std::string>(v);                   // throws std::bad_variant_access:
the wrong alternative
if (auto* s = std::get_if<std::string>(&v)) { }   // a pointer, or nullptr:
no exception
v = std::string("hi");                      // now index 1; the int is gone
std::variant<int, std::string> d;           // default: holds a value-initialised
FIRST alternative (index 0, int 0)
```

Visiting every alternative

`std::visit` calls a function with whichever alternative is active. Give it a callable that handles **every** alternative, and a missing one is a **compile error**, not a silent bug.

```
template <class... Ts> struct overloaded : Ts... { using Ts::operator()...; };
// glue lambdas into one callable
template <class... Ts> overloaded(Ts...) -> overloaded<Ts...>;
// deduction guide (needed before C++20)

struct Circle { double r; };
struct Rect   { double w, h; };
using Shape = std::variant<Circle, Rect>;          // no base class, no virtual functions

double area(const Shape& s) {
    return std::visit(overloaded{
        [](const Circle& c) { return 3.0 * c.r * c.r; },
        [](const Rect& r)   { return r.w * r.h; }
    }, s);
}
std::vector<Shape> shapes{Circle{2}, Rect{3, 4}};   // a heterogeneous collection stored by value
// total area: 12 + 12 = 24 (compiled)
```

- **Variant or virtual functions?** A variant fits a *closed* set of types you control (messages, tokens, syntax-tree nodes), where you keep adding operations. Virtual functions fit an *open* set that others may extend. The trade-off is spelled out under Visitor in More Patterns.
- If an alternative's constructor throws during assignment, the variant can end up *valueless by exception* (`valueless_by_exception()`). It is rare, and it is why you should keep alternatives cheap and non-throwing to move.
- A reference alternative is ill-formed (use `std::reference_wrapper`). The same type twice is legal, but then you can only reach it by index, not by type.
- `std::monostate` is an empty alternative: put it first to make a variant default-constructible when no alternative should be the default.

`expected`: a value or an error

`std::expected<T, E>` (C++23) holds either the result `T` or an error `E`. It is `optional` that explains *why* the value is missing, and an alternative to exceptions for failures that are normal.

```
enum class ParseError { Empty, NotANumber };

std::expected<int, ParseError> parse(std::string_view s) {
    if (s.empty()) return std::unexpected(ParseError::Empty);        // the
error path
    if (auto n = parseInt(s)) return *n;                             // the
value path
    return std::unexpected(ParseError::NotANumber);
}

auto r = parse("12");
if (r) use(*r);                                  // true: it holds a value
else   report(r.error());                        // otherwise: the error
parse("zz").value_or(0);                         // 0: a default when it holds
an error
parse("").value();                               // throws
std::bad_expected_access<ParseError>
```

FAILURE STYLE	COST AND VISIBILITY	USE FOR
Exception	invisible in the signature; a throw is slow	rare failures the caller cannot handle here, constructors
Error code plus out-parameter	easy to ignore, clumsy to write	C APIs
`optional<T>`	visible, free, says nothing about why	"not found", where the reason is obvious
`expected<T, E>`	visible, free, carries the reason	expected failures: parsing, I/O, validation

It needs a C++23 standard library (GCC 12 or later, recent MSVC). With an older one, `std::variant<T, E>` or a small result struct does the same job, and the free `tl::expected` library has the same interface. Mark the function `[[nodiscard]]` so the result cannot be ignored.

`span` and `string_view`: non-owning views

A **view** is a pointer plus a length: it looks at elements that someone else owns. `std::span<T>` (C++20) views any contiguous run of `T`; `std::string_view` (C++17) is the same idea for characters.

```
long sum(std::span<const int> xs) { long t = 0; for (int x : xs) t += x;
return t; }

std::vector<int> vec{1, 2, 3, 4, 5};
int arr[3] = {10, 20, 30};
sum(vec);                                // 15: a vector converts to a span
with no copy
sum(arr);                                // 60: so does a C array, and its size
is remembered
sum(std::span(vec).subspan(1, 3));       // 9:  elements 2, 3, 4: a slice is
just a new pointer and length
sum(std::span(vec).first(2));            // 3
```

- A span is two words (compiled: 16 bytes on a 64-bit machine). Pass it **by value**.
- `span<const int>` is read-only; `span<int>` lets the function modify the elements in place. It still cannot grow or shrink the container.
- It replaces the `(T* data, size_t n)` pair. Overflows are easier to spot, and a function written against `span` accepts a vector, an array, a `std::array` or a part of any of them.
- **Dangling.** A view must not outlive what it looks at. A `span` of a vector that then reallocates (`push_back`), or a `string_view` of a temporary string, points at freed memory. Use views for parameters and short-lived local use, not for members that live longer than the data.

> RULE OF THUMB Take a view to *read or modify* existing elements; take or return the owning type (`vector`, `string`) to *keep* them.

Which one when

SITUATION	USE
Return two related values that are used right away	`pair` / structured bindings; a named `struct` if they travel further
A lookup that may find nothing	`optional<T>`
A parse or I/O call whose failure has a reason	`expected<T, E>` (or an exception if it is rare)
A value that is one of a few known types	`variant` plus `visit`
A function that reads a run of elements	`span<const T>`; `string_view` for text
A value of any type, unknown at compile time	`std::any` (rare: you lose compile-time checking; prefer a variant)

The same idea runs through all of them: make illegal or missing states **explicit in the type** so the compiler, not a comment, enforces them.

Further reading by section

Where each section of this lesson is explained in more depth. Page numbers are PDF viewer pages.

pair, tuple and structured bindings

C++ Primer (Lippman, Lajoie, Moo)

- 17.1 The tuple type (PDF pages 747 to 751)

variant and std::visit

C++ Primer (Lippman, Lajoie, Moo)

- 19.5 to 19.6 Nested classes and unions (PDF pages 872 to 880)